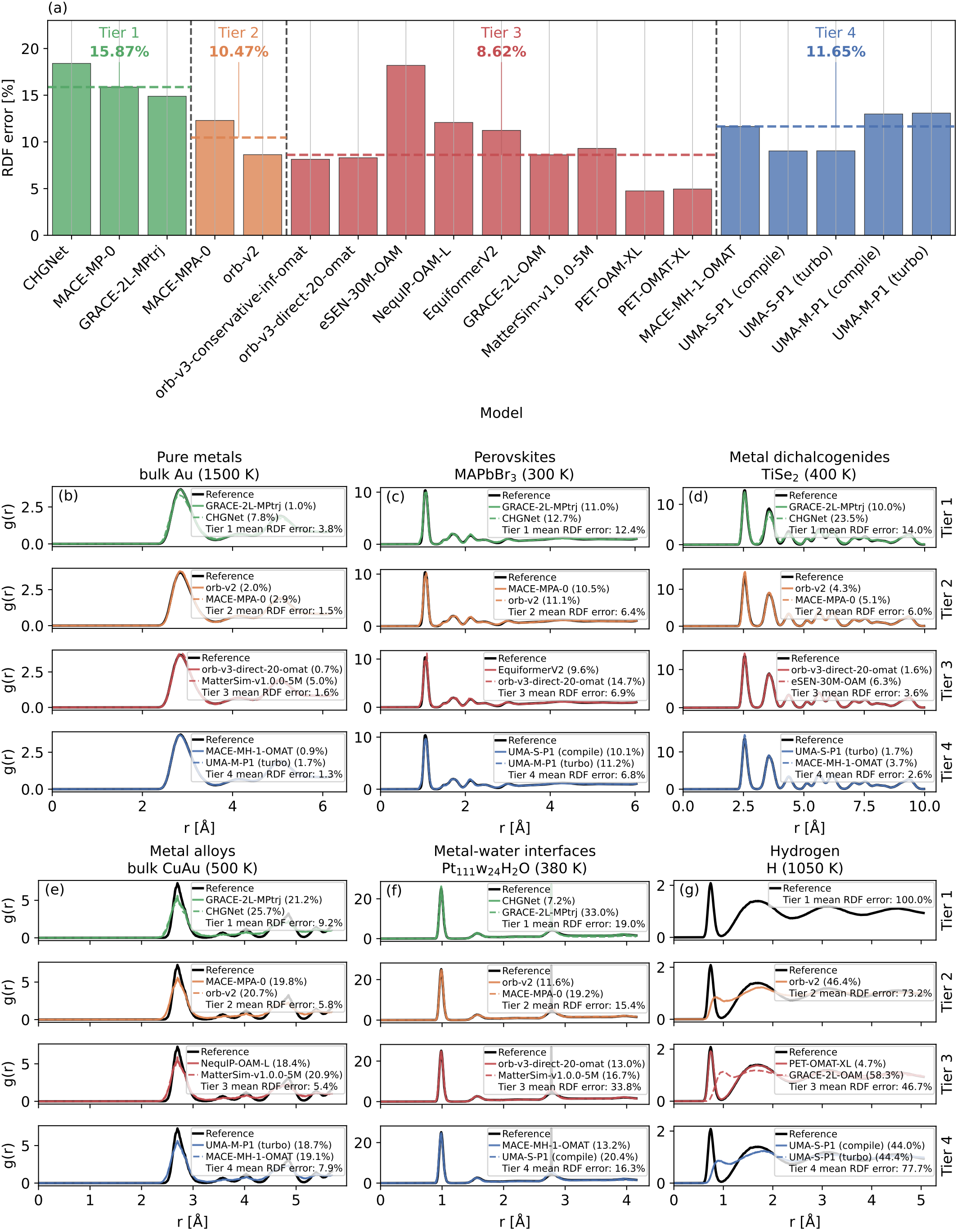

(a)
Tier 1
15.87%
Tier 2
10.47%
Tier 3
8.62%
Tier 4
11.65%
RDF error [%]
0
5
10
15
20
CHGNet
MACE-MP-0
GRACE-2L-MPtrj
MACE-MPA-0
orb-v2
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
eSEN-30M-OAM
NequIP-OAM-L
EquiformerV2
GRACE-2L-OAM
MatterSim-v1.0.0-5M
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1 (compile)
UMA-S-P1 (turbo)
UMA-M-P1 (compile)
UMA-M-P1 (turbo)
Model
Pure metals
bulk Au (1500 K)
(b)
Reference
GRACE-2L-MPtrj (1.0%)
CHGNet (7.8%)
Tier 1 mean RDF error: 3.8%
orb-v2 (2.0%)
MACE-MPA-0 (2.9%)
Tier 2 mean RDF error: 1.5%
orb-v3-direct-20-omat (0.7%)
MatterSim-v1.0.0-5M (5.0%)
Tier 3 mean RDF error: 1.6%
MACE-MH-1-OMAT (0.9%)
UMA-M-P1 (turbo) (1.7%)
Tier 4 mean RDF error: 1.3%
Perovskites
MAPbBr3 (300 K)
(c)
GRACE-2L-MPtrj (11.0%)
CHGNet (12.7%)
Tier 1 mean RDF error: 12.4%
MACE-MPA-0 (10.5%)
orb-v2 (11.1%)
Tier 2 mean RDF error: 6.4%
EquiformerV2 (9.6%)
orb-v3-direct-20-omat (14.7%)
Tier 3 mean RDF error: 6.9%
UMA-S-P1 (compile) (10.1%)
UMA-M-P1 (turbo) (11.2%)
Tier 4 mean RDF error: 6.8%
Metal dichalcogenides
TiSe2 (400 K)
(d)
GRACE-2L-MPtrj (10.0%)
CHGNet (23.5%)
Tier 1 mean RDF error: 14.0%
orb-v2 (4.3%)
MACE-MPA-0 (5.1%)
Tier 2 mean RDF error: 6.0%
orb-v3-direct-20-omat (1.6%)
eSEN-30M-OAM (6.3%)
Tier 3 mean RDF error: 3.6%
UMA-S-P1 (turbo) (1.7%)
MACE-MH-1-OMAT (3.7%)
Tier 4 mean RDF error: 2.6%
Metal alloys
bulk CuAu (500 K)
(e)
GRACE-2L-MPtrj (21.2%)
CHGNet (25.7%)
Tier 1 mean RDF error: 9.2%
MACE-MPA-0 (19.8%)
orb-v2 (20.7%)
Tier 2 mean RDF error: 5.8%
NequIP-OAM-L (18.4%)
MatterSim-v1.0.0-5M (20.9%)
Tier 3 mean RDF error: 5.4%
UMA-M-P1 (turbo) (18.7%)
MACE-MH-1-OMAT (19.1%)
Tier 4 mean RDF error: 7.9%
Metal-water interfaces
Pt111W24H2O (380 K)
(f)
CHGNet (7.2%)
GRACE-2L-MPtrj (33.0%)
Tier 1 mean RDF error: 19.0%
orb-v2 (11.6%)
MACE-MPA-0 (19.2%)
Tier 2 mean RDF error: 15.4%
orb-v3-direct-20-omat (13.0%)
MatterSim-v1.0.0-5M (16.7%)
Tier 3 mean RDF error: 33.8%
MACE-MH-1-OMAT (13.2%)
UMA-S-P1 (compile) (20.4%)
Tier 4 mean RDF error: 16.3%
Hydrogen
H (1050 K)
(g)
Tier 1 mean RDF error: 100.0%
orb-v2 (46.4%)
Tier 2 mean RDF error: 73.2%
PET-OMAT-XL (4.7%)
GRACE-2L-OAM (58.3%)
Tier 3 mean RDF error: 46.7%
UMA-S-P1 (compile) (44.0%)
UMA-S-P1 (turbo) (44.4%)
Tier 4 mean RDF error: 77.7%
g(r)
r [Å]
Tier 1
Tier 2
Tier 3
Tier 4